

1 Solution — GIAM 5.3

Problem 12

1.1 Problem

Give an inductive proof that $\forall n \geq 3 \in \mathbb{N}, n^3 + 3 > n^2 + 3n + 1$.

1.2 Answer

$$D(k+1) = D(k) + 3k^2 + k - 3, \quad D(n) = (n^3 + 3) - (n^2 + 3n + 1)$$

The gap starts positive at the basis, $D(3) = 30 - 19 = 11$, and the increment $3k^2 + k - 3$ is positive for every $k \geq 1$. Verified for every n from 3 to 5000.

1.3 The Proof

Write $D(n) = (n^3 + 3) - (n^2 + 3n + 1) = n^3 - n^2 - 3n + 2$. The claim is that $D(n) > 0$ for $n \geq 3$.

Theorem. For every natural number $n \geq 3$, $n^3 + 3 > n^2 + 3n + 1$.

Proof. By induction on n , starting at 3.

Basis. At $n = 3$, $3^3 + 3 = 30$ and $3^2 + 9 + 1 = 19$, and $30 > 19$.

Inductive step. Let $k \geq 3$ be a particular but arbitrarily chosen natural number and suppose $D(k) > 0$. Expanding,

$$\begin{aligned} D(k+1) &= (k+1)^3 - (k+1)^2 - 3(k+1) + 2 \\ &= (k^3 + 3k^2 + 3k + 1) - (k^2 + 2k + 1) - 3k - 3 + 2 \\ &= (k^3 - k^2 - 3k + 2) + 3k^2 + k - 3 \\ &= D(k) + 3k^2 + k - 3. \end{aligned}$$

Since $k \geq 1$, $3k^2 + k - 3 \geq 3 + 1 - 3 = 1 > 0$. Hence $D(k+1) > D(k) > 0$, which says $(k+1)^3 + 3 > (k+1)^2 + 3(k+1) + 1$.

Basis and step together give the result for every $n \geq 3$. ■

As in Problems 10 and 11, the step uses less than it is given: $k \geq 1$ suffices. The basis is what pins the theorem to $n \geq 3$.

1.4 The Picture

The gap factors, and its sign pattern explains every small case.

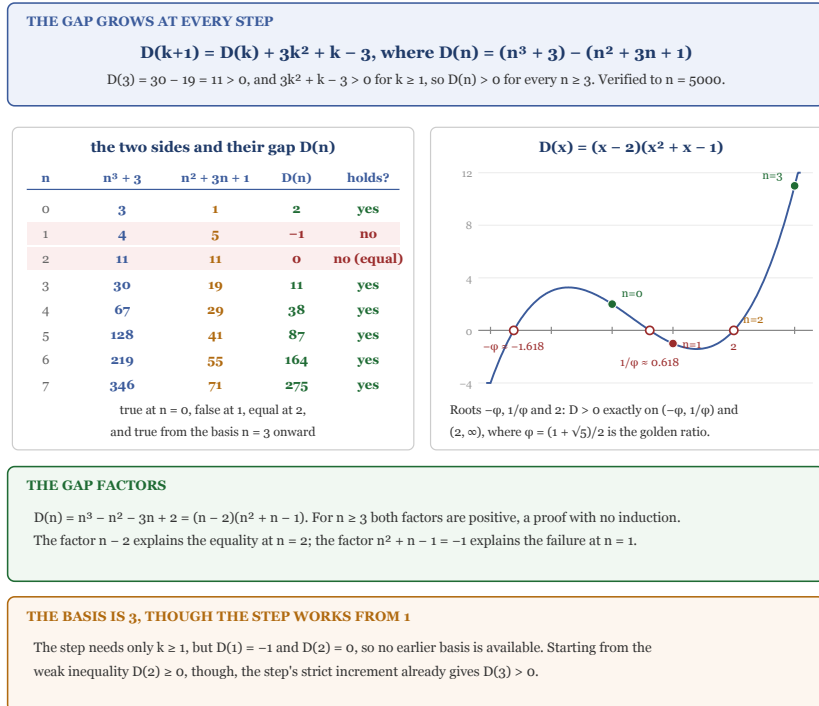


Figure 1.1: A diagram in four parts. The top band gives the step identity D of k plus one equals D of k plus three k squared plus k minus three, with D of 3 equal to 11. The left panel tabulates both sides and their gap for n from zero to seven: the inequality holds at n equal to zero, fails at one, is an equality at two, and holds from three on. The right panel graphs D of x, the product of x minus two and x squared plus x minus one, with roots at minus the golden ratio, its reciprocal, and two, and the integer points zero to three marked. A green band gives the factorisation, and an orange band notes that the step works from k equal to one while the basis must be three.

1.5 Remark — The Gap Factors

Since $D(2) = 8 - 4 - 6 + 2 = 0$, the number 2 is a root, and dividing out $n - 2$ gives

$$D(n) = n^3 - n^2 - 3n + 2 = (n - 2)(n^2 + n - 1),$$

checked for $|n|$ up to 1000. For $n \geq 3$ both factors are positive, which proves the theorem without induction. It also explains the small cases: the factor $n - 2$ gives equality at $n = 2$; at $n = 1$ the factors are -1 and 1 , so the inequality fails; at $n = 0$ both are negative, so it holds.

The quadratic factor has roots $\frac{-1 \pm \sqrt{5}}{2}$, that is, $1/\varphi \approx 0.618$ and $-\varphi \approx -1.618$, where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. So over the reals the inequality holds exactly on $(-\varphi, 1/\varphi) \cup (2, \infty)$. The golden ratio appears because $x^2 + x - 1 = 0$ is its defining equation in disguise: substituting $x = 1/y$ gives $y^2 = y + 1$.

1.6 Remark — The Basis, and a Weaker Starting Point

The step is valid from $k = 1$, yet neither $n = 1$ (where $D = -1$) nor $n = 2$ (where $D = 0$) can serve as the basis of the strict inequality. But the step adds at least 1, so it upgrades a weak inequality to a strict one: from $D(2) \geq 0$ it gives $D(3) \geq 0 + 3 \cdot 4 + 2 - 3 = 11 > 0$. So an induction for $D(n) \geq 0$ from the basis $n = 2$ yields the strict result for $n \geq 3$.

1.7 Remark — Problems 10 to 12 Side by Side

The last three problems share one pattern: a polynomial or exponential inequality that fails for small n , a basis at the first n where it holds, and a step that compares how fast the two sides grow. In each the step works earlier than the basis does, so the basis decides where the theorem begins.

problem	fails at	basis	step valid for
10 : $3n^2 + 3n + 1 < 2n^3$	0, 1, 2	3	$k \geq 1$
11 : $n^3 < 3^n$	3	4	$k \geq 3$
12 : $n^3 + 3 > n^2 + 3n + 1$	1, 2	3	$k \geq 1$